

POLS 641

Introductory Political Analysis

Class 3: Basics of Probability

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Northern Illinois University

September 8, 2026

Today's Agenda

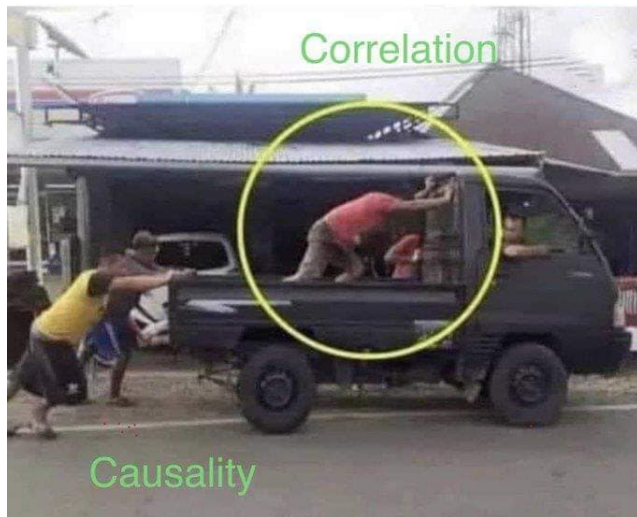
- Final Project Questions
- Causality Review
- Probability
 - Key Terms
 - Conditional Probability
 - Distributions

Some Course Logistics

- Datacamp due tomorrow - next datacamp due in two weeks
- Problem set due next Wednesday.
- Posted on BB and Website. Due on BB
- Knit as you go

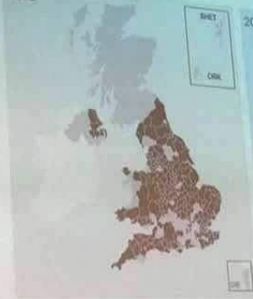
Questions on Final Project?

Review: Correlation and Causation



1992 – MAD COW DISEASE OUTBREAK AREAS

1992



City

■ BSE-Areas ■ BSE-Free-Areas

2016 – BREXIT REFERENDUM LOCAL RESULTS

2016



City

■ Majority Leave ■ Majority Remain

IT MAY, HOWEVER, BE A MISTAKE TO JUMP TO CONCLUSIONS

Review: Endogeneity Another way of saying correlation $\neq$ causation

- Endogeneity: The IV is correlated with the error term.
 - ‘Confounder:’ This means that there is another unmeasured variable (a confounder) that affects the IV which also affects the DV.
 - We haven’t included this other confounding variable in our model.
 - Consequently, calling our observed correlation causal could be ‘spurious’ – it’s really a different variable, which we haven’t measured, that’s driving the results.
- To get our causal estimate, we need to create exogeneity:
 - We need our IV to be unrelated to (uncorrelated with) the error term.
 - We can do this through random assignment

Review: Our Core Model

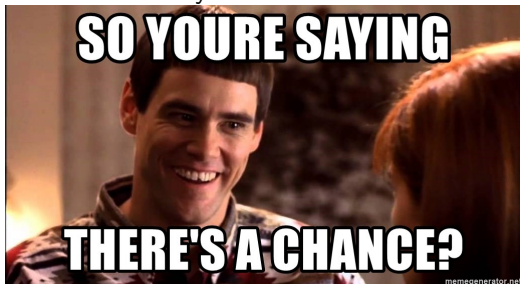
$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

- Y_i = dependent variable, outcome variable, the thing we want to predict
- X_i = independent variable, explanatory variable, the thing that predicts the DV
- ϵ_i = the error term, the part of the DV our IV doesn't explain, everything NOT in our model (usually a lot!).
Note: the error term is not directly observable.

Review of Key Terms

- 1 Independent Variable
- 2 Dependent Variable
- 3 Correlation
- 4 Causation
- 5 Endogeneity
- 6 Randomization
- 7 Probabilistic
- 8 Core Statistical Model

Questions on causality? the core model? The course?



Probability

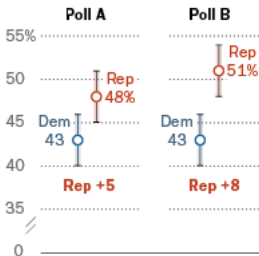


Probability

For election polls, different measures of the race have different margins of error

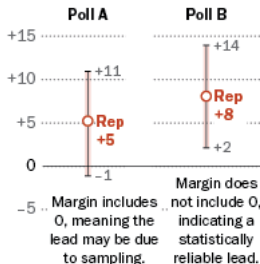
The margin of error reported for most polls applies to support for individual candidates ...

Margin of error for single candidate support
(MOE $\pm$ 3 pct. points)



... while the margin of error for a candidate's lead is nearly twice as large.

Margin of error for difference between two candidates' level of support (%Rep - %Dem)
(MOE $\pm$ 6 pct. points)



Probability

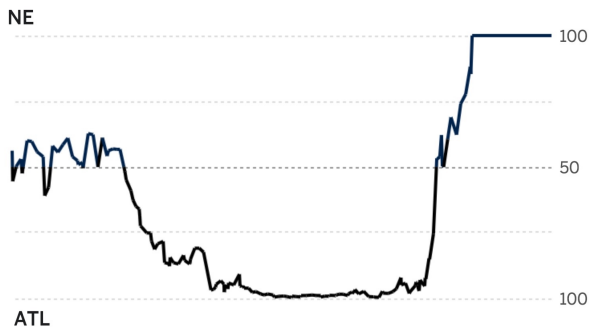


Probability



Probability

Win Probability



Some Probability Theory

Today we'll start building a more theoretical foundation that we need in order to go farther.

One of the hardest things about statistics is understanding *distributions* of outcomes: the idea that you *could* observe different outcomes with different probabilities, even when you only observe one outcome.

Statistics has to do with understanding properties of that *distribution* of potential outcomes, not just the data you have.

Today we'll work on some *probability theory* and repeatedly think about distributions of potential outcomes.

Almost all interesting political phenomena are probabilistic

Probability

- Necessary when analyzing situations characterized by "chance" or "randomness" i.e. data-generating process that can realize any number of different values.
- a random process is a process that can be repeated many times with different outcomes each time
- Goal today: very basic introduction to ideas of probability theory
- **As social scientists, we see the world in probabilistic rather than deterministic ways**

In every day life, we often encounter the use of the term probability. It is used in many different ways. For example:

- 1 The probability that a fair coin will land heads is $1/2$.
- 2 The probability that a selection of 6 numbers wins the National Lottery Lotto jackpot is 1 in $\binom{19}{6} = 13983816$ or 7.15112×10^{-8} .
- 3 The probability that a water bottle will land 'point up' is 0.62.
- 4 The probability that a large earthquake will occur on the San Andreas Fault in the next 30 years is about 21%
- 5 The probability that humanity will be extinct by 2100 is about 50%

The first two cases are things derived from logic. For example, we know that the coin either lands heads or tails. By definition, a fair coin is equally likely to land heads or tail. So the probability of either must be $1/2$. aka classical probability

The third is something probability derived from experiments. Perhaps we did 1000 experiments and 620 of the bottles point up. aka frequentist
The fourth and fifth examples belong to yet another category that talks about are beliefs and predictions. aka subjective probability

Terminology

- Sample Space: Ω ("omega") the set of all possible outcomes of a process.
- Events: A subset of the sample space to which we assign probabilities
- The **probability** of an event is the proportion of times the outcome would occur if we observed the random process an infinite number of times(*Law of Large Numbers*).

Probability Axioms

- Nonnegativity: P is greater is always greater than 0
- Normalization: $P(\Omega) = 1$
- Additivity: Can add probabilities(as long as mutually exclusive/disjoint)

Defining Probability

- i.e. Flipping a coin once: $\Omega = (H, T)$
- Flipping a coin twice: $\Omega = (HH, HT, TH, TT)$
- $P(\cdot)$ encodes knowledge or belief about the likelihood of each outcome
- A probability function assigns a probability to each outcome in Ω :
 $P(\cdot) \rightarrow (0, 1)$

Defining Probability

- i.e. A vote in congress: $\Omega = (\text{Yes}, \text{No}, \text{Abstain})$
- A probability function assigns a probability to each outcome in Ω :
 $P(\text{Yes}) \rightarrow (0,1)$
- What is an Ω for party identification in the US be?

Core idea: Independence

Events A and B are independent if knowing the outcome A provides no information about whether B occurred. $Pr(A|B) = Pr(A)$

Note: symmetric - true going both ways.

We often assume independence:

- An individual's opinion is independent of their neighbors
- A country's decision to ratify a treaty is independent of what other countries are doing

*Often a problem when measuring a single unit at multiple points in time.

Multiplication Rule

- probability of a joint outcome is the multiplication of the probabilities of each individual outcome
- $P(A \text{ and } B) = P(A) * P(B)$
- $P(\text{two heads in a row}) = P(\text{heads the first time}) * P(\text{heads the second time}) = .5 * .5 = .25$

Conditional Probability

Conditional probability is the likelihood of an event occurring given occurrence of another event

$$P(A \cap B) = P(A, B) = P(B) \times P(A|B)$$

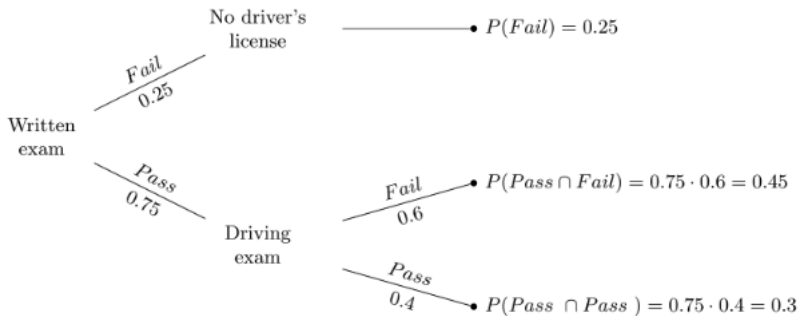
Definition 1: Probability of event A given that event B has occurred.

Example: Probability of war between two countries given that they are both democracies.

Probability of vote for a candidate given gender or SES

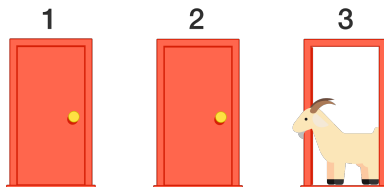
Visualizing Conditional Probability

Probability Tree



Conditional Probability

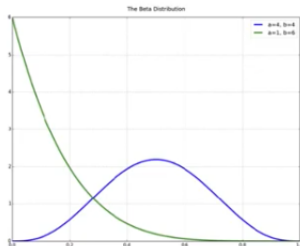
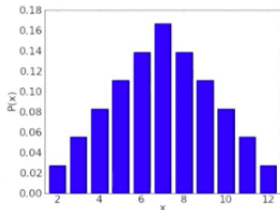
The Monty Hall Problem



Probability Density Function

Distribution - the range of possible values for a random variable and the associated relative probabilities for each value.

- The outcomes listed must be disjoint.
- Each probability must be between 0 and 1.
- The probabilities must total 1.

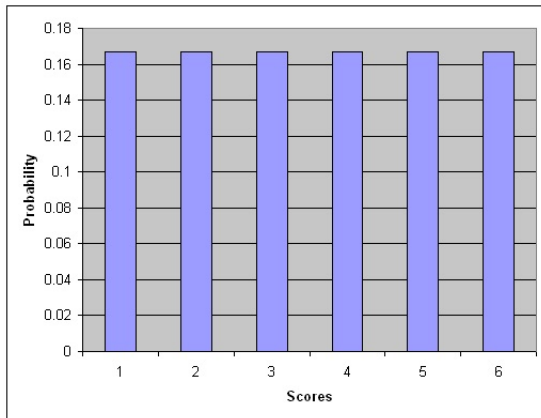


discrete and

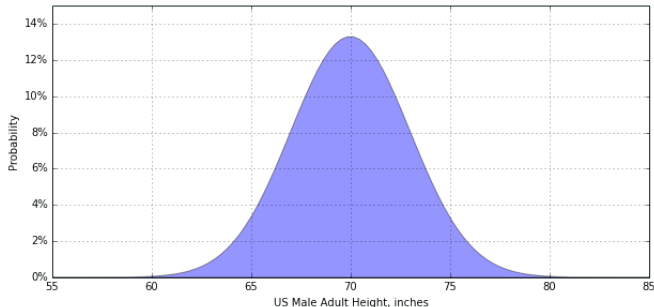
continuous

Discrete Probability Density Function

PDF of Dice

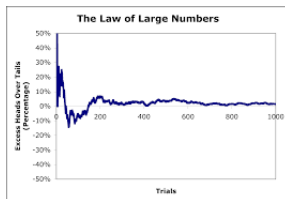


Continuous Probability Density Function



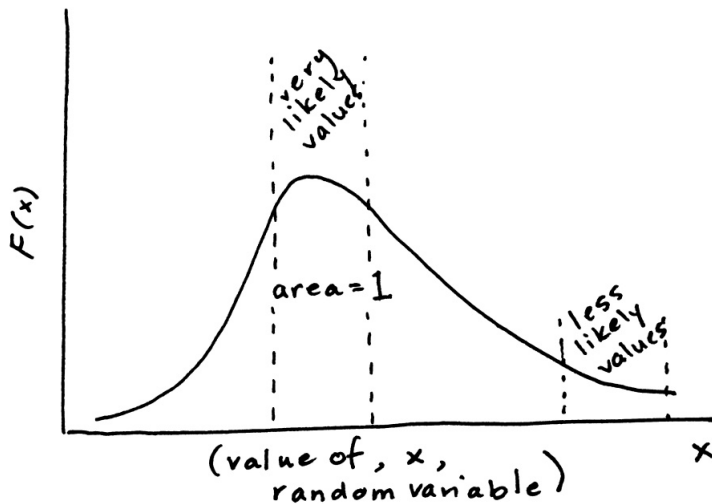
How do we calculate probability of a particular height?(aka a specific event)

The Law of Large Numbers

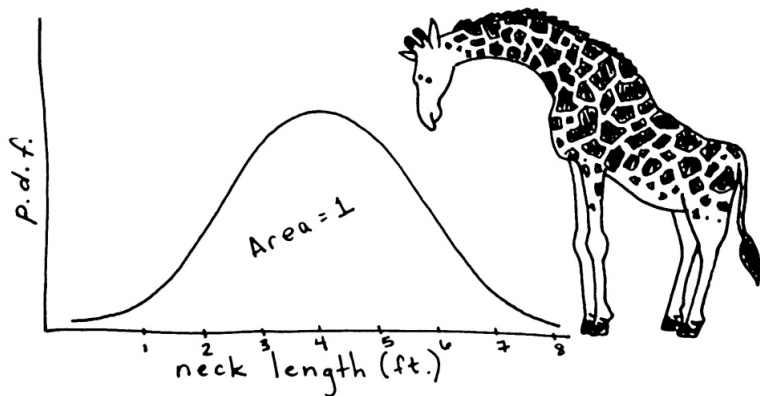


As more observations are collected, the proportion of occurrences with a particular outcome converges to the probability p of that outcome. Using a frequentist approach we can define probabilities based on what we've observed.

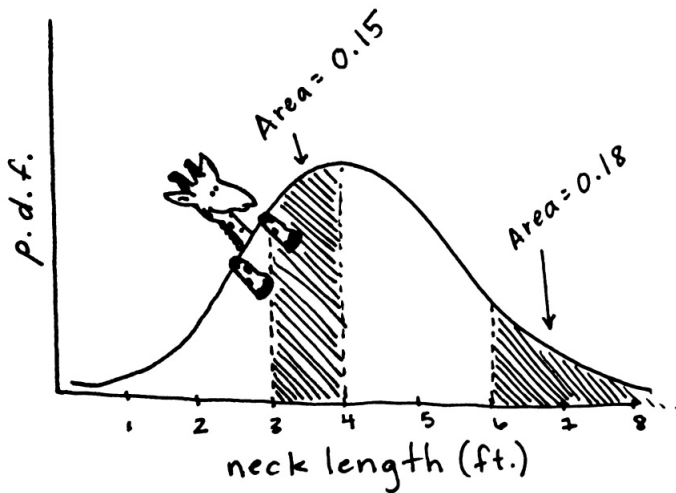
Reading a Probability Density Function



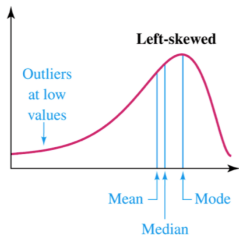
Reading a Probability Density Function



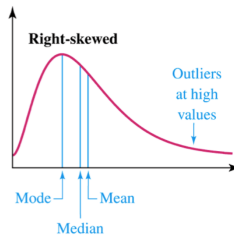
Reading a Probability Density Function



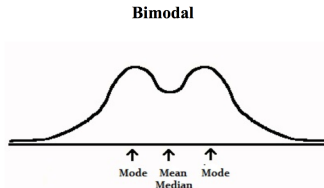
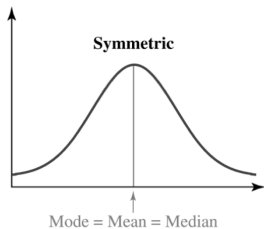
Reading a Probability Density Function - Means



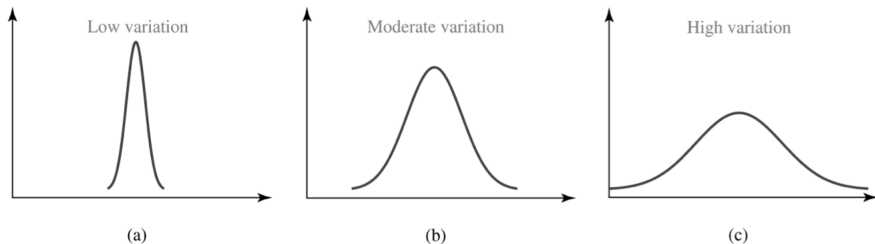
(a)



(b)



Reading a Probability Density Function - Variance



Frequentist vs Bayesian

Two major philosophies about the nature of probability

- **Frequentist:** From repeated sampling. ie the percentage of the time that some event occurs if the data generating process is repeated over and over again. This course and most political science
- **Bayesian:** From prior knowledge. ie judgement given about how plausible and outcome is in a given situation, conditioned by prior knowledge and assumptions.

Example: Coin Flipping $\Omega=(H, T)$ Frequentist: $\# \text{ of } H / \# \text{ of Trials}$

Bayesian: what prior knowledge they have about the coin, and make a judgement about its likelihood of coming up heads or tails. They would then update their estimated probability by observing the results of their experiment (or realization of a data-generating process).

Bottom Line: We often assume independence, but its not always a fair assumption.

I.E. Assume voting preferences are NOT conditional on the preferences of others.

Can be accounted for using different models

Big Ideas

- an understanding of probability theory helps us understand how what we observe relates to the real world.
- statistics has to with understanding properties of the distribution of potential outcomes, not just the data you have
- idea that you can observe different outcomes with different probabilities, even when you only observe one outcome

Course Logistics

Problem set # 1 Due next Wednesday, at Midnight
Turn in Rmarkdown (.rmd) and knitted doc(.html)

Reminder

My Office Hours: Tuesday 10-11, Thursday 330-430 (Zulauf 410) and
Online



Before spending \$ at datacamp, please check out: [#DumpDataCamp](#)

Next Time: Random variables

Read: Chapter 2 only pages 49-64 (45-60 in first edition)
No RIA for next week

